Pennsylvania PSSA Science Assessment Analysis (2008-2024)

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Introduction

This analysis examines over 15 years of Pennsylvania State System of Assessment (PSSA) science test scores to understand how student science proficiency has evolved over time. We explore trends across four proficiency levels—Advanced, Proficient, Basic, and Below Basic—to identify patterns that can inform educational policy and highlight areas where students may need additional support.

Data Overview

The dataset includes statewide PSSA science assessment results from the 2008-09 school year through the 2023-24 school year. Note that data is missing for 2013-14 and the 2019-21 period (due to COVID-19 disruptions). Each year's data shows the percentage of Pennsylvania students in each of the four proficiency categories.

Setup and Data Preparation

First, we load the necessary libraries and create our dataset from the PSSA historical data.

```
# PSSA Science Assessment Analysis
# Pennsylvania State System of Assessment (2008-2024)
# Authors: Ritvik Kothapalyam & Amal Parekh

### Summary Statistics Here we calculate the mean, minimum, maximum, and standard deviation

# Load required libraries
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.1.4     v readr     2.1.5
v forcats   1.0.1     v stringr   1.5.2
v ggplot2   4.0.0     v tibble    3.3.0
v lubridate 1.9.4     v tidyr     1.3.1
v purrr     1.1.0
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becom
```

```r
library(ggplot2)

# Create the dataset manually from the PDF data
pssa_data <- data.frame(
  LocationType = rep("State", 52),
  Location = rep("Pennsylvania", 52),
  Score = c(
    # 2008-09
    "Advanced", "Proficient", "Basic", "Below Basic",
    # 2009-10
    "Advanced", "Proficient", "Basic", "Below Basic",
    # 2010-11
    "Advanced", "Proficient", "Basic", "Below Basic",
    # 2011-12
    "Advanced", "Proficient", "Basic", "Below Basic",
    # 2012-13
    "Advanced", "Proficient", "Basic", "Below Basic",
    # 2014-15
    "Advanced", "Proficient", "Basic", "Below Basic",
    # 2015-16
    "Advanced", "Proficient", "Basic", "Below Basic",
    # 2016-17
    "Advanced", "Proficient", "Basic", "Below Basic",
    # 2017-18
    "Advanced", "Proficient", "Basic", "Below Basic",
    # 2018-19
    "Advanced", "Proficient", "Basic", "Below Basic",
    # 2021-22
    "Advanced", "Proficient", "Basic", "Below Basic",
    # 2022-23
    "Advanced", "Proficient", "Basic", "Below Basic",
```

```r
    # 2023-24
    "Advanced", "Proficient", "Basic", "Below Basic"
  ),
  TimeFrame = c(
    rep("2008-09", 4), rep("2009-10", 4), rep("2010-11", 4), rep("2011-12", 4),
    rep("2012-13", 4), rep("2014-15", 4), rep("2015-16", 4), rep("2016-17", 4),
    rep("2017-18", 4), rep("2018-19", 4), rep("2021-22", 4), rep("2022-23", 4),
    rep("2023-24", 4)
  ),
  DataFormat = rep("Percent", 52),
  Data = c(
    # 2008-09
    0.272, 0.319, 0.245, 0.164,
    # 2009-10
    0.280, 0.314, 0.231, 0.175,
    # 2010-11
    0.279, 0.330, 0.235, 0.157,
    # 2011-12
    0.278, 0.337, 0.241, 0.144,
    # 2012-13
    0.34487, 0.34694, 0.14985, 0.15834,
    # 2014-15
    0.339, 0.339, 0.152, 0.170,
    # 2015-16
    0.33418, 0.3356, 0.14443, 0.18529,
    # 2016-17
    0.271, 0.366, 0.213, 0.150,
    # 2017-18
    0.281, 0.366, 0.215, 0.138,
    # 2018-19
    0.309, 0.371, 0.195, 0.125,
    # 2021-22
    0.285, 0.337, 0.195, 0.183,
    # 2022-23
    0.307, 0.349, 0.182, 0.163,
    # 2023-24
    0.295, 0.363, 0.181, 0.161
  )
)

# Convert Data to percentage (multiply by 100)
pssa_data$Percentage <- pssa_data$Data * 100
```

```
# Set factor levels for Score to maintain proper order
pssa_data$Score <- factor(pssa_data$Score,
                          levels = c("Advanced", "Proficient", "Basic", "Below Basic"))

# Create a year variable for easier plotting
pssa_data$Year <- as.numeric(substr(pssa_data$TimeFrame, 1, 4))

# Display the cleaned data
print("PSSA Science Assessment Data (2008-2024)")
```

```
[1] "PSSA Science Assessment Data (2008-2024)"
```

```
print(head(pssa_data, 10))
```

```
   LocationType     Location       Score TimeFrame DataFormat  Data Percentage
1         State Pennsylvania    Advanced   2008-09    Percent 0.272       27.2
2         State Pennsylvania  Proficient   2008-09    Percent 0.319       31.9
3         State Pennsylvania       Basic   2008-09    Percent 0.245       24.5
4         State Pennsylvania Below Basic   2008-09    Percent 0.164       16.4
5         State Pennsylvania    Advanced   2009-10    Percent 0.280       28.0
6         State Pennsylvania  Proficient   2009-10    Percent 0.314       31.4
7         State Pennsylvania       Basic   2009-10    Percent 0.231       23.1
8         State Pennsylvania Below Basic   2009-10    Percent 0.175       17.5
9         State Pennsylvania    Advanced   2010-11    Percent 0.279       27.9
10        State Pennsylvania  Proficient   2010-11    Percent 0.330       33.0
   Year
1  2008
2  2008
3  2008
4  2008
5  2009
6  2009
7  2009
8  2009
9  2010
10 2010
```

```
# Summary statistics Here we calculate the mean, minimum, maximum, and standard deviation fo

cat("\n=== SUMMARY STATISTICS ===\n")
```

```
=== SUMMARY STATISTICS ===
```

```r
summary_stats <- pssa_data %>%
  group_by(Score) %>%
  summarize(
    Mean_Percentage = mean(Percentage),
    Min_Percentage = min(Percentage),
    Max_Percentage = max(Percentage),
    SD_Percentage = sd(Percentage)
  )
print(summary_stats)
```

```
# A tibble: 4 x 5
  Score       Mean_Percentage Min_Percentage Max_Percentage SD_Percentage
  <fct>                 <dbl>          <dbl>          <dbl>         <dbl>
1 Advanced               29.8           27.1           34.5          2.64
2 Proficient             34.4           31.4           37.1          1.83
3 Basic                  19.8           14.4           24.5          3.52
4 Below Basic            16.0           12.5           18.5          1.73
```

```r
# Calculate combined proficiency rates (Advanced + Proficient)
proficiency_rates <- pssa_data %>%
  filter(Score %in% c("Advanced", "Proficient")) %>%
  group_by(TimeFrame, Year) %>%
  summarize(Combined_Proficiency = sum(Percentage), .groups = "drop")

cat("\n=== COMBINED PROFICIENCY RATES (Advanced + Proficient) ===\n")
```

```
=== COMBINED PROFICIENCY RATES (Advanced + Proficient) ===
```

```r
print(proficiency_rates)
```

```
# A tibble: 13 x 3
   TimeFrame  Year Combined_Proficiency
   <chr>     <dbl>                <dbl>
 1 2008-09    2008                 59.1
 2 2009-10    2009                 59.4
 3 2010-11    2010                 60.9
```

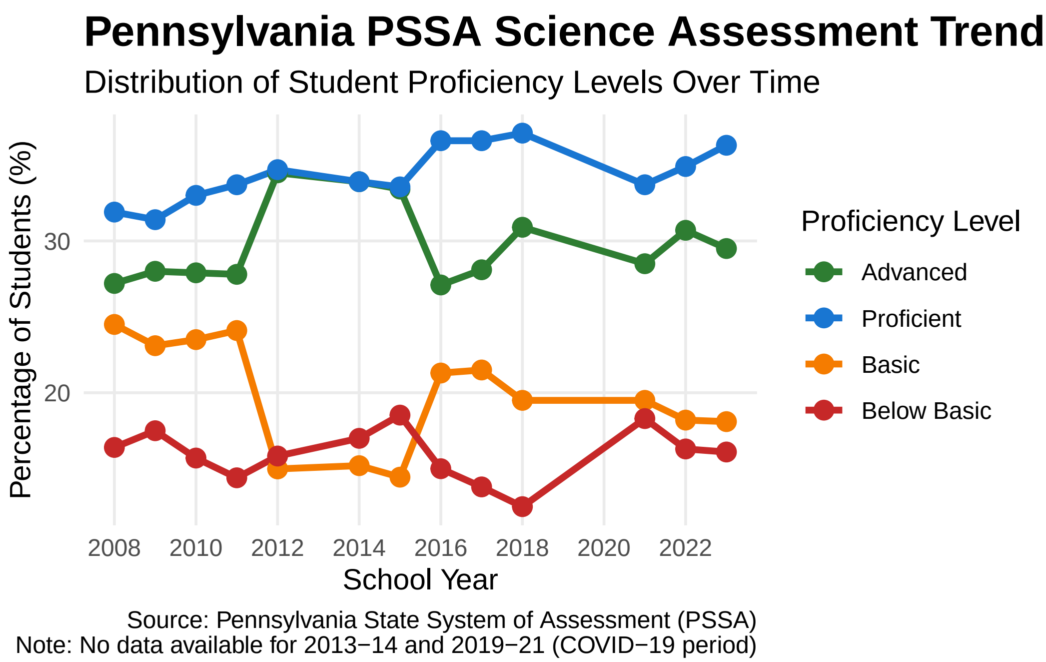
```
 4 2011-12    2011                   61.5
 5 2012-13    2012                   69.2
 6 2014-15    2014                   67.8
 7 2015-16    2015                   67.0
 8 2016-17    2016                   63.7
 9 2017-18    2017                   64.7
10 2018-19    2018                   68
11 2021-22    2021                   62.2
12 2022-23    2022                   65.6
13 2023-24    2023                   65.8
```

```r
# GRAPH 1: Line Chart showing trends over time
print("Creating Graph 1: Line Chart of All Proficiency Levels...")
```

```
[1] "Creating Graph 1: Line Chart of All Proficiency Levels..."
```

```r
ggplot(pssa_data, aes(x = Year, y = Percentage, color = Score, group = Score)) +
  geom_line(size = 1.2) +
  geom_point(size = 3) +
  scale_color_manual(values = c("Advanced" = "#2E7D32",
                                "Proficient" = "#1976D2",
                                "Basic" = "#F57C00",
                                "Below Basic" = "#C62828")) +
  labs(
    title = "Pennsylvania PSSA Science Assessment Trends (2008-2024)",
    subtitle = "Distribution of Student Proficiency Levels Over Time",
    x = "School Year",
    y = "Percentage of Students (%)",
    color = "Proficiency Level",
    caption = "Source: Pennsylvania State System of Assessment (PSSA)\nNote: No data availab
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(size = 16, face = "bold"),
    plot.subtitle = element_text(size = 12),
    legend.position = "right",
    panel.grid.minor = element_blank()
  ) +
  scale_x_continuous(breaks = seq(2008, 2023, 2))
```

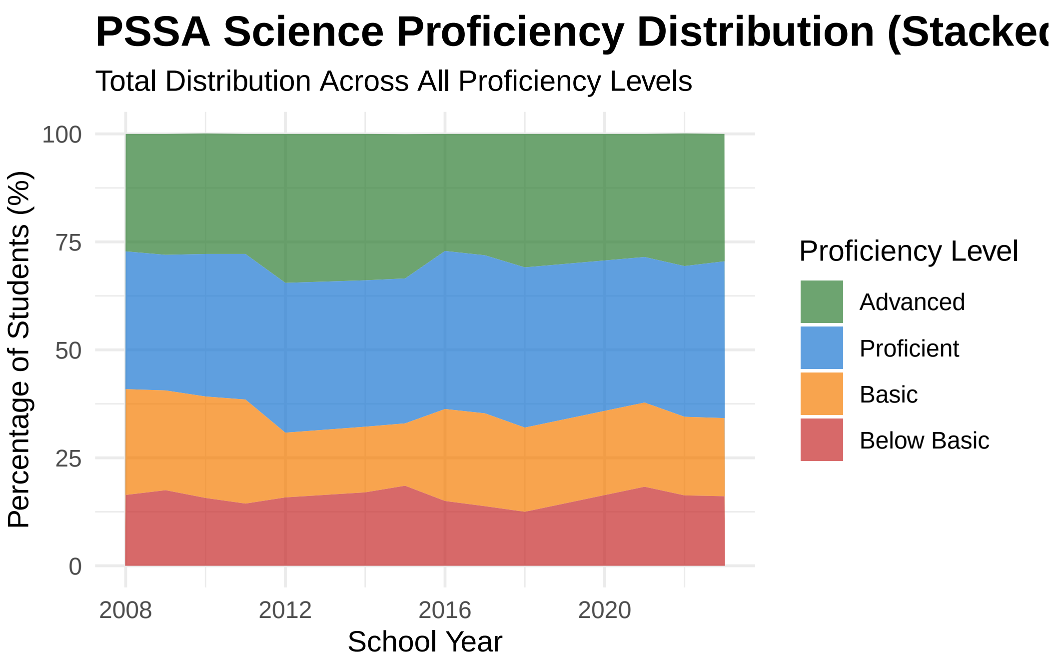
```
Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
i Please use `linewidth` instead.
```




```r
# GRAPH 2: Stacked Area Chart
print("Creating Graph 2: Stacked Area Chart...")
```

```
[1] "Creating Graph 2: Stacked Area Chart..."
```

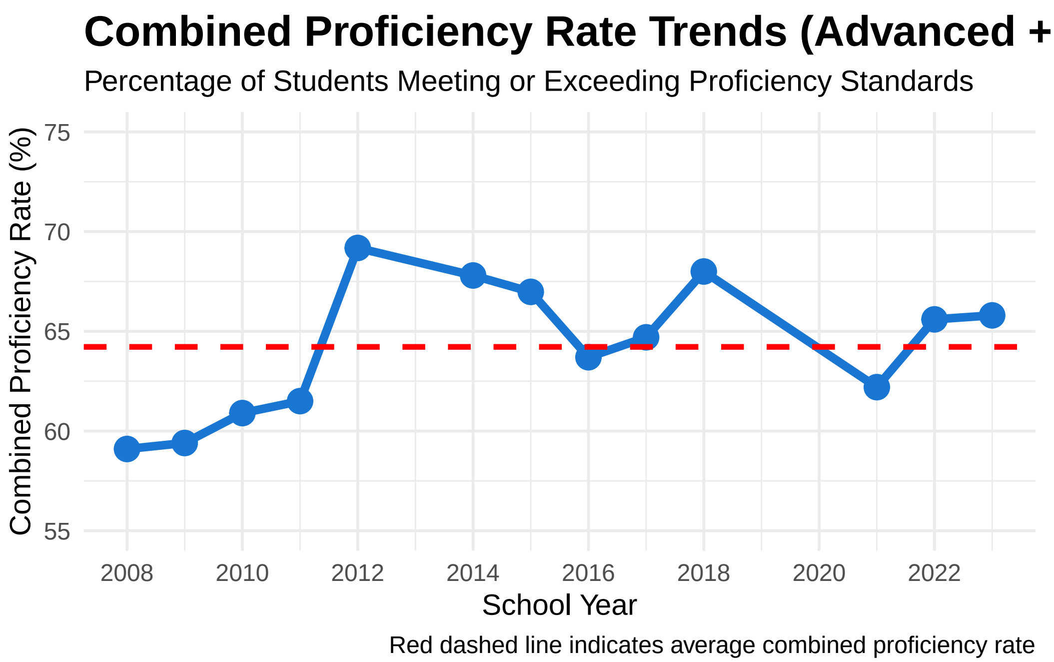
```r
ggplot(pssa_data, aes(x = Year, y = Percentage, fill = Score)) +
  geom_area(alpha = 0.7) +
  scale_fill_manual(values = c("Advanced" = "#2E7D32",
                               "Proficient" = "#1976D2",
                               "Basic" = "#F57C00",
                               "Below Basic" = "#C62828")) +
  labs(
    title = "PSSA Science Proficiency Distribution (Stacked)",
    subtitle = "Total Distribution Across All Proficiency Levels",
    x = "School Year",
    y = "Percentage of Students (%)",
    fill = "Proficiency Level"
  ) +
  theme_minimal() +
  theme(plot.title = element_text(size = 16, face = "bold"))
```




```r
# GRAPH 3: Combined Proficiency Rate Chart
print("Creating Graph 3: Combined Proficiency Rate...")
```

```
[1] "Creating Graph 3: Combined Proficiency Rate..."
```

```r
ggplot(proficiency_rates, aes(x = Year, y = Combined_Proficiency)) +
  geom_line(color = "#1976D2", size = 1.5) +
  geom_point(color = "#1976D2", size = 4) +
  geom_hline(yintercept = mean(proficiency_rates$Combined_Proficiency),
             linetype = "dashed", color = "red", size = 1) +
  labs(
    title = "Combined Proficiency Rate Trends (Advanced + Proficient)",
    subtitle = "Percentage of Students Meeting or Exceeding Proficiency Standards",
    x = "School Year",
    y = "Combined Proficiency Rate (%)",
    caption = "Red dashed line indicates average combined proficiency rate"
  ) +
  theme_minimal() +
  theme(plot.title = element_text(size = 16, face = "bold")) +
  scale_x_continuous(breaks = seq(2008, 2023, 2)) +
  ylim(55, 75)
```




```
cat("\n=== ANALYSIS COMPLETE ===\n")
```

```
=== ANALYSIS COMPLETE ===
```

```
cat("Three graphs have been created!\n")
```

```
Three graphs have been created!
```

```
cat("Look at the 'Plots' pane (bottom-right) to view them.\n")
```

```
Look at the 'Plots' pane (bottom-right) to view them.
```

```
cat("Use the arrow buttons to navigate between graphs.\n")
```

```
Use the arrow buttons to navigate between graphs.
```

```
cat("Click 'Export' to save graphs as images.\n")
```

```
Click 'Export' to save graphs as images.
```

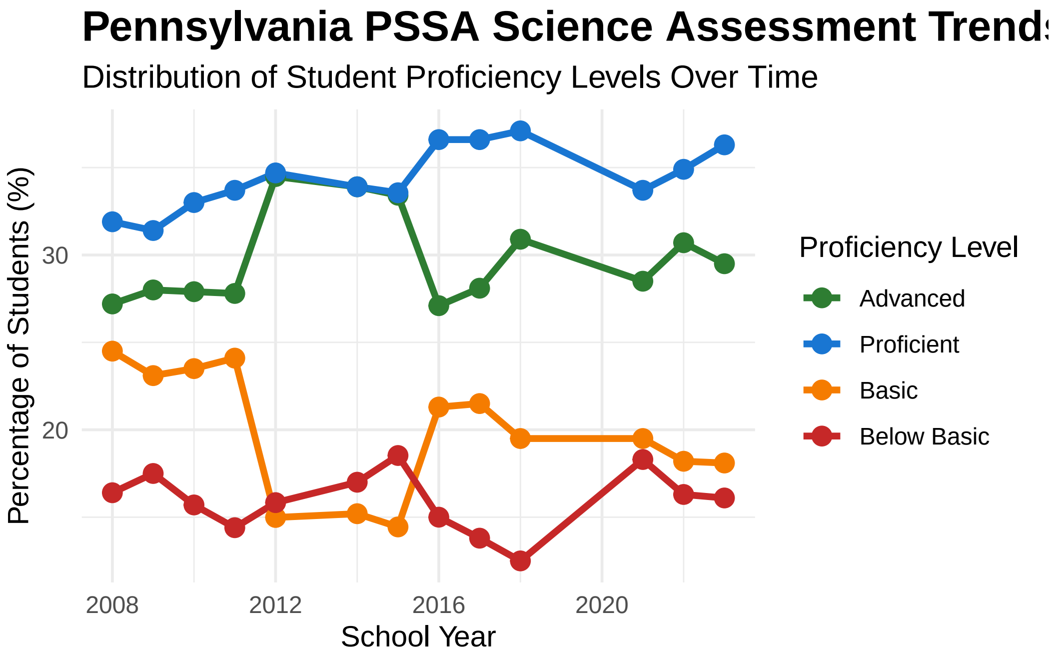
You can add options to executable code like this

```r
# Console Commands
# Convert Data to percentage (multiply by 100)
pssa_data$Percentage <- pssa_data$Data * 100

# Set factor levels for Score to maintain proper order
pssa_data$Score <- factor(pssa_data$Score,
                          levels = c("Advanced", "Proficient", "Basic", "Below Basic"))

# Create a year variable for easier plotting
pssa_data$Year <- as.numeric(substr(pssa_data$TimeFrame, 1, 4))

# GRAPH 1: Line Chart showing trends over time
ggplot(pssa_data, aes(x = Year, y = Percentage, color = Score, group = Score)) +
  geom_line(linewidth = 1.2) +
  geom_point(size = 3) +
  scale_color_manual(values = c("Advanced" = "#2E7D32",
                                 "Proficient" = "#1976D2",
                                 "Basic" = "#F57C00",
                                 "Below Basic" = "#C62828")) +
  labs(
    title = "Pennsylvania PSSA Science Assessment Trends (2008-2024)",
    subtitle = "Distribution of Student Proficiency Levels Over Time",
    x = "School Year",
    y = "Percentage of Students (%)",
    color = "Proficiency Level"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(size = 16, face = "bold"),
    plot.subtitle = element_text(size = 12),
    legend.position = "right"
  )
```



Pennsylvania PSSA Science Assessment Trends
Distribution of Student Proficiency Levels Over Time
Percentage of Students (%)
30
20
2008
2012
2016
2020
School Year
Proficiency Level
Advanced
Proficient
Basic
Below Basic